SPARSE KOOPMAN NEURAL OPERATOR FOR ATMOSPHERIC DYNAMICS WITH SPECTRAL STABILITY AND LATENT SCALING

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We introduce *Sparse Koopman Neural Operator (SKNO)* [4] for atmospheric field prediction, motivated by the need for stable, interpretable, and low-dimensional representations of geophysical dynamics. A few contemporary aspects of sparse modeling includes LASSO [6] or graph-based sparse neural networks [5], to name a few. Koopman operator methods have shown strong predictive performance but the structure and scaling of their latent dynamics are still poorly understood. Our proposed model combines a neural lifting map with a sparse latent Koopman operator [1] and a decoder for reconstructing the next atmospheric field. The learned latent dynamics are linear in Koopman space [3], while sparsity promotes structured and interpretable mode interactions. We evaluate the approach on real atmospheric temperature data and analyze both one-step prediction and multi-step rollout behavior. We also study the Koopman spectrum, operator sparsity, and how these quantities change with latent dimension. The learned Koopman operators are spectrally stable, with spectral radius remaining ≈ 1, and the resulting multi-step rollouts remain stable over time. Increasing the latent Koopman dimension improves both one-step and rollout performance: in our experiments, the final loss decreases from approximately 0.330 at latent dimension 16 to 0.217 at latent dimension 128, while the 5-step rollout error decreases from approximately 0.442 to 0.194. At the same time, Koopman sparsity increases moderately, and gains begin to saturate beyond roughly 64 latent modes. This suggests that the atmospheric dynamics admit a relatively low effective Koopman rank despite the higher-dimensional state representation. These results indicate that SKNO provides a promising framework for scientific machine learning on atmospheric fields: they yield stable and interpretable latent dynamics, accurate short and medium-horizon forecasts, and a natural way to study latent scaling through Koopman spectral structure. Our experiments suggest that model capacity in operator learning may be governed not only by network size, but also by the spectral compressibility of the underlying dynamics.

1 Methodology and Result

Given an input field $x_t \in \mathbb{R}^d$, a neural encoder maps the state to a latent representation $z_t = \mathbf{E}(x_t) \in \mathbb{R}^n$. The latent dynamics are governed by a linear Koopman operator $\mathcal{K} \in \mathbb{R}^{n \times n}$, such that $z_{t+1} = \mathcal{K} z_t$. A decoder reconstructs the next state as $\hat{x}_{t+1} = \mathbf{D}(z_{t+1})$. We promote interpretability and structured latent dynamics by imposing sparsity on $\mathcal{K}$ through soft-thresholding.

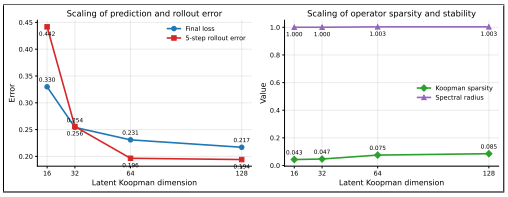


Figure 1: **SKNO scaling behavior on atmospheric data.**

In Figure 1 result: **(left)** one-step prediction loss and 5-step rollout error decrease as latent Koopman dimension increases, with diminishing returns beyond 64 modes; **(right)** Koopman operator sparsity and spectral radius remain stable across scales. The spectral radius stays close to one, indicating stable latent dynamics.

References

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A Algorithm for SKNO with soft-thresholding sparsity

Algorithm 1 Sparse Koopman Neural Operator

Require: Training pairs (x_t, x_{t+1}), encoder $\mathbf{E}_{\theta_E}$, decoder $\mathbf{D}_{\theta_D}$, latent dimension n, sparsity threshold τ, learning rate η, loss-balance hyperparameter $\lambda > 0$

Initialize encoder parameters θ_E, decoder parameters θ_D
Initialize Koopman operator $\mathcal{K} \in \mathbb{R}^{n \times n}$
for each training epoch **do**
  for each batch (x_t, x_{t+1}) **do**
    Encode: $z_t \leftarrow \mathbf{E}_{\theta_E}(x_t)$
    Koopman evolution: $z_{t+1} \leftarrow \mathcal{K} z_t$
    Decode: $\hat{x}_{t+1} \leftarrow \mathbf{D}_{\theta_D}(z_{t+1})$
    Prediction loss:

$$\mathcal{L}_{\text{pred}} \leftarrow \|\hat{x}_{t+1} - x_{t+1}\|_2^2$$

    Latent consistency loss:

$$\mathcal{L}_{\text{Koop}} \leftarrow \|z_{t+1} - \mathbf{E}_{\theta_E}(x_{t+1})\|_2^2$$

    Total loss:

$$\mathcal{L} \leftarrow \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{Koop}}$$

    Gradient update:

$$(\theta_E, \theta_D, \mathcal{K}) \leftarrow (\theta_E, \theta_D, \mathcal{K}) - \eta \nabla \mathcal{L}$$

    Sparsity enforcement (soft-thresholding):

$$\mathcal{K}_{ij} \leftarrow \text{sign}(\mathcal{K}_{ij}) \cdot \max(|\mathcal{K}_{ij}| - \tau, 0)$$

  end for
end for
return Trained encoder $\mathbf{E}_{\theta_E}$, decoder $\mathbf{D}_{\theta_D}$, sparse Koopman operator $\mathcal{K}$

B Experimental Setup

We evaluate the proposed SKNO model on a real atmospheric temperature dataset (accessed via the `xarray` [2] repository) represented on a 25×53 spatial grid. Each sample consists of a flattened spatial field, and the task is to predict the next time step. The model uses a neural encoder and decoder, with a latent Koopman dimension varying from 16 to 128. The Koopman operator is learned jointly with the encoder and decoder, with sparsity enforced via thresholding. Models are trained using mean squared error loss for 10 epochs. We report one-step prediction loss, multi-step rollout error (5-step), Koopman sparsity, and spectral radius.

C Additional Experimental Details

We provide additional qualitative and spectral analysis of the learned Koopman dynamics. The results demonstrate stable latent evolution, structured sparsity, and consistent scaling behavior across model sizes.

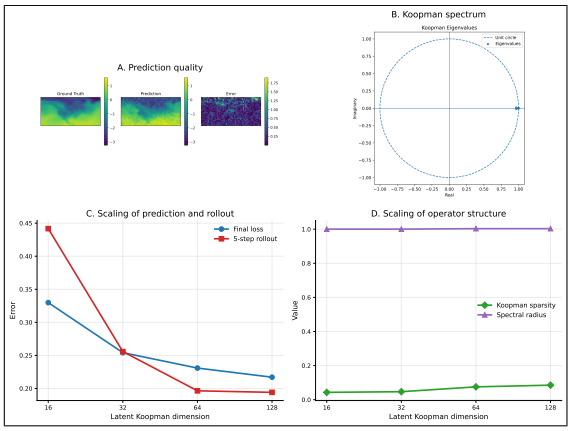


Figure 2: **Comprehensive evaluation of SKNO on atmospheric data.** (A) Ground truth, prediction, and error showing accurate reconstruction of large-scale structures. (B) Koopman eigenvalues lie close to the unit circle, indicating stable latent dynamics. (C) Prediction and rollout errors decrease with latent dimension, with diminishing returns beyond approximately 64 modes. (D) Koopman operators remain sparse while maintaining spectral radius near one.

Code availability

Code will be made publicly available on corresponding author HS* github profile `https://himanshuvnm.github.io/` upon publication.